

Exercise 2 - SQL (Aggregate Functions & grouping)

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Data Analytics - March 2026

1. **SELECT COUNT (*) AS** total - number - of - employees
FROM employees;

total - number - of - employees
10

2. **SELECT SUM (salary) AS** total - salary
FROM employees
WHERE department = 'IT';

total - salary
220 000

3. **SELECT AVG (salary) AS** average - salary
FROM employees
WHERE department = 'HR';

average - salary
49 500

4. **SELECT MIN (salary) AS** lowest - salary
MAX (salary) AS highest - salary
FROM employees;

lowest - salary	highest - salary
48 000	62 000

5. SELECT department, SUM(salary) AS total-salary

FROM employees

GROUP BY department;

department	total-salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6. SELECT city, COUNT(*) AS total-number-of-employees

FROM employees

GROUP BY city;

city	total-number-of-employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. SELECT department, AVG(salary) AS average-salary

FROM employees

GROUP BY department

ORDER BY average-salary DESC;

department	average-salary
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

8. **SELECT** department, **SUM**(salary) **AS** total-salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100 000;

department	total-salary
IT	220 000
Finance	119 000
Marketing	105 000

9. **SELECT** city, **COUNT**(*) **AS** total-number-of-employees
FROM employees
GROUP BY city
HAVING COUNT(*) > 1
ORDER BY total-number-of-employees **DESC**;

city	total-number-of-employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. **SELECT** department, **AVG**(salary) **AS** average-salary
FROM employees
GROUP BY department
ORDER BY average-salary **DESC**
Limit 1;

department	average-salary
Finance	59 500

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